A Human-Centric MARL Framework for On-Ramp Merging in Mixed-Autonomous Traffic

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Introduction

The Reality

- The near-future is mixed-autonomy traffic, where Connected and Automated Vehicles (CAVs) make up only 5% to 20% [CAV penetration rate] of vehicles.

The Issue

- Most traditional research assume either full observability and/or a high CAV penetration rate.
- When CAV PR becomes low, these models fail, triggering hard brakes or causing stop and go traffic.

The Idea

- We formulate the problem as a sparse control problem.
- Can a small fraction of cooperative CAVs help guide and improve human-dominated traffic flow?
- CAVs act as "Traffic Shepherds"; they help shape local interactions to improve general traffic dynamics instead of dominating the system.

The Approach

- We propose a human-centric multi-agent reinforcement learning (MARL) framework where CAVs learn high level interaction roles.
- The goal is to shift importance from individual CAV efficiency to improving overall traffic flow.

Methodology

Environment.

- 2-lane main highway with an on-ramp and merging lane simulated via SUMO and TraCI.
- Mixture of HDVs: cautious (green), normal (white), and aggressive (blue), and CAVs (red)

Human-Centric Reward Function

- Five weights balancing efficiency, safety, stability, fairness, and control cost.

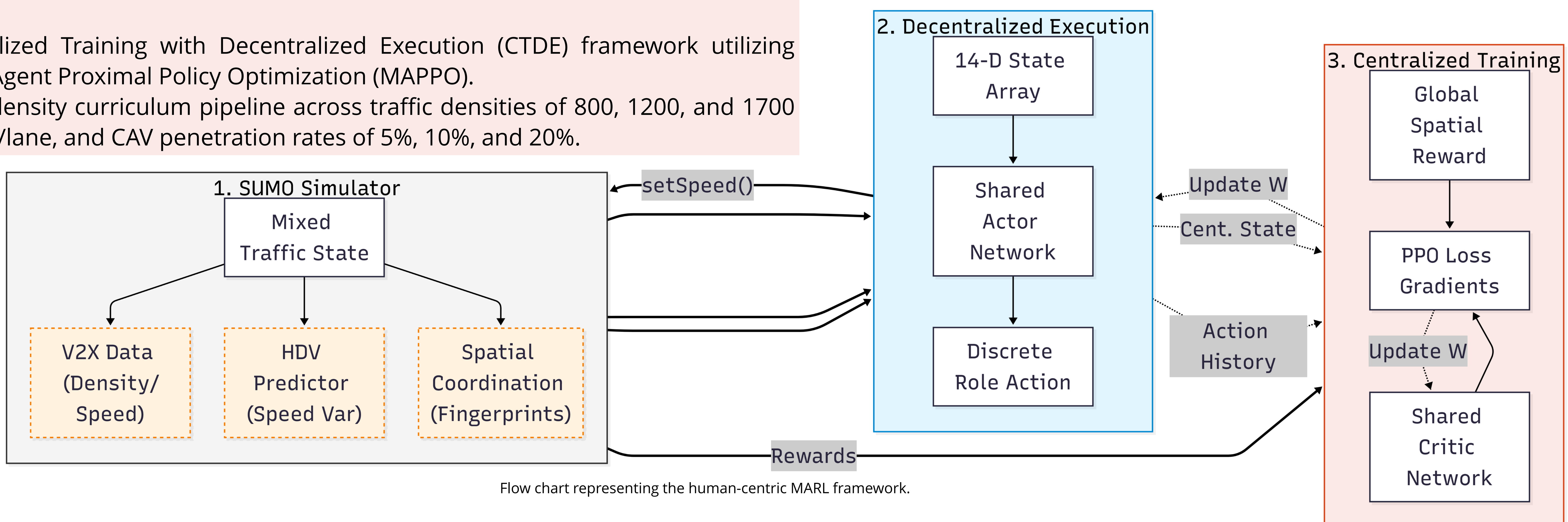
$$r_i = w_{\text{eff}} \cdot r_{\text{eff}} + w_{\text{safe}} \cdot r_{\text{safe}} + w_{\text{stab}} \cdot r_{\text{stab}} + w_{\text{fair}} \cdot r_{\text{fair}} - w_{\text{ctrl}} \cdot c_{\text{ctrl}}$$

Evaluation

- Tested in different traffic densities (800 and 1700 veh/hr/lane) with varying CAV penetration rates (5%, 10%, 20%).
- Data is recorded and compared against other policies (All-HDV, Rule-CAV, Selfish-CAV, Flat-MARL).

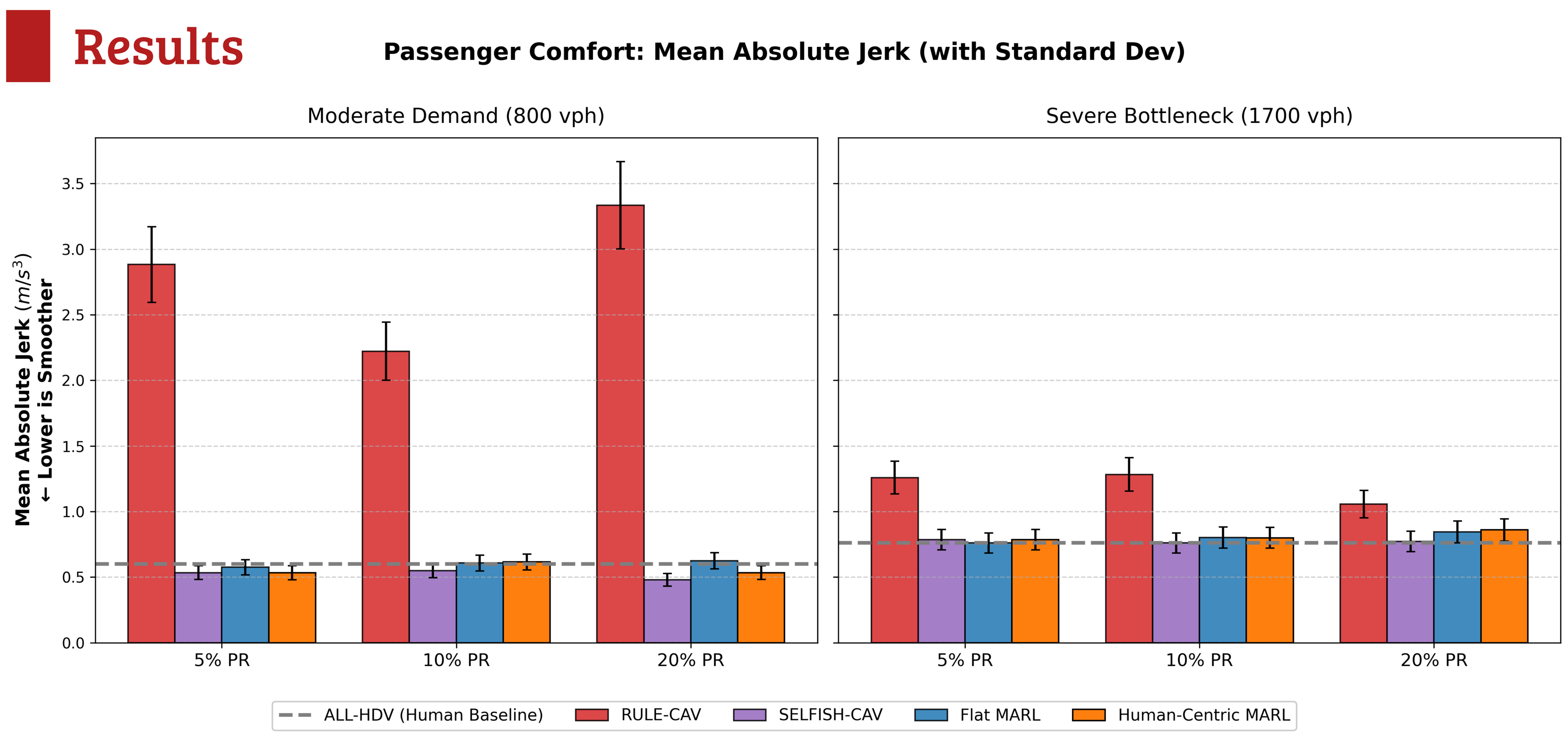
Training

- Centralized Training with Decentralized Execution (CTDE) framework utilizing Multi-Agent Proximal Policy Optimization (MAPPO).
- Multi-density curriculum pipeline across traffic densities of 800, 1200, and 1700 veh/hr/lane, and CAV penetration rates of 5%, 10%, and 20%.

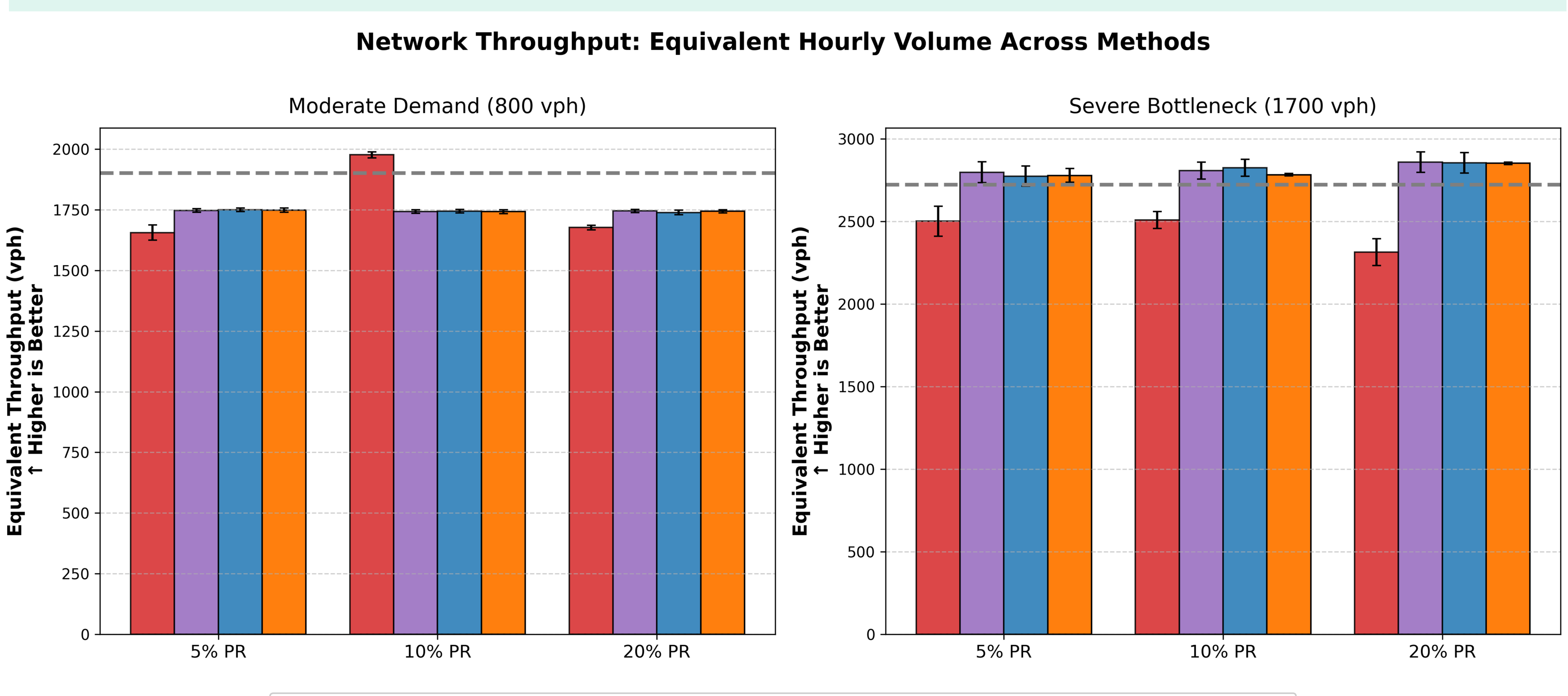


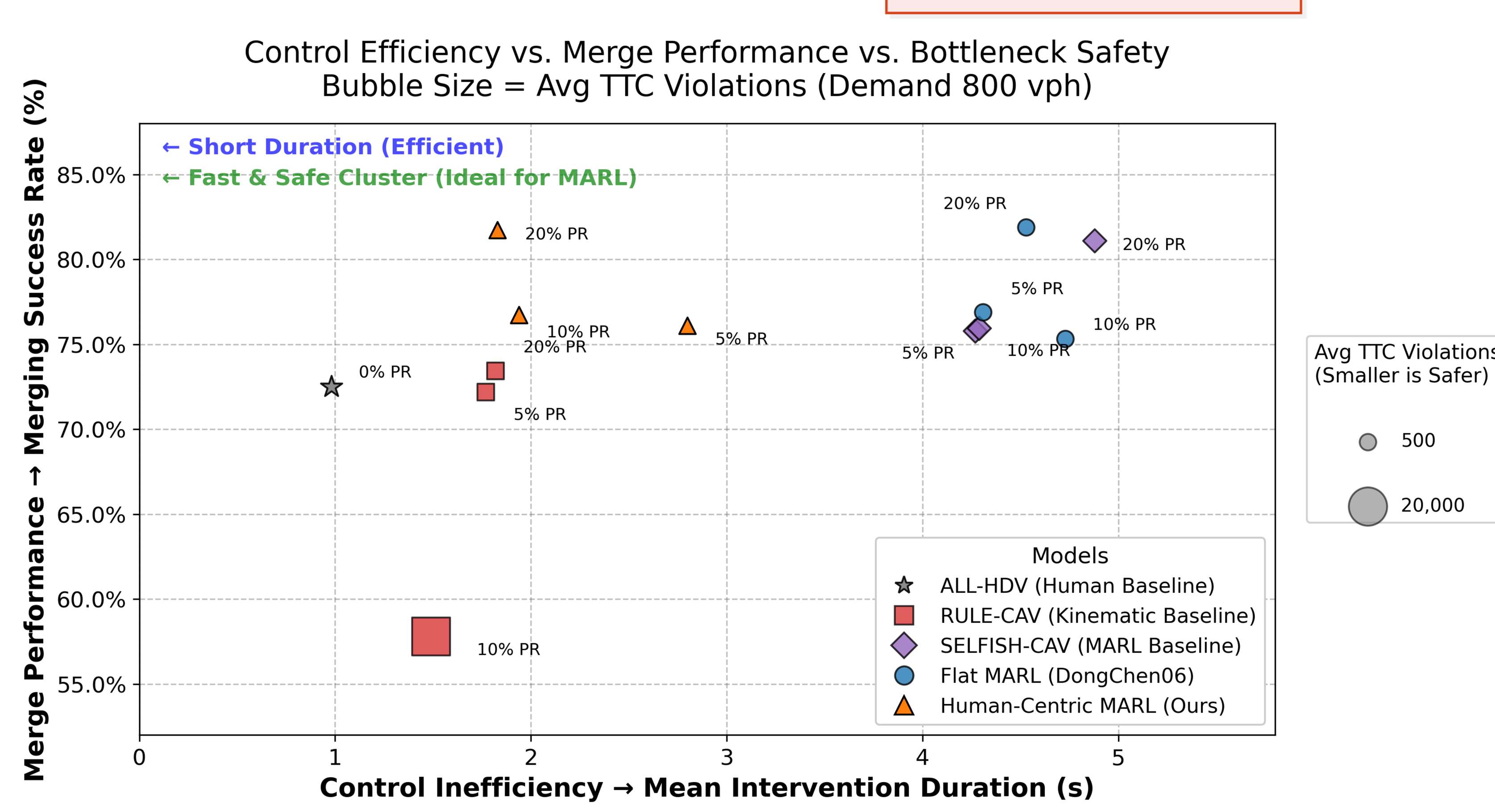
Flow chart representing the human-centric MARL framework.

Results

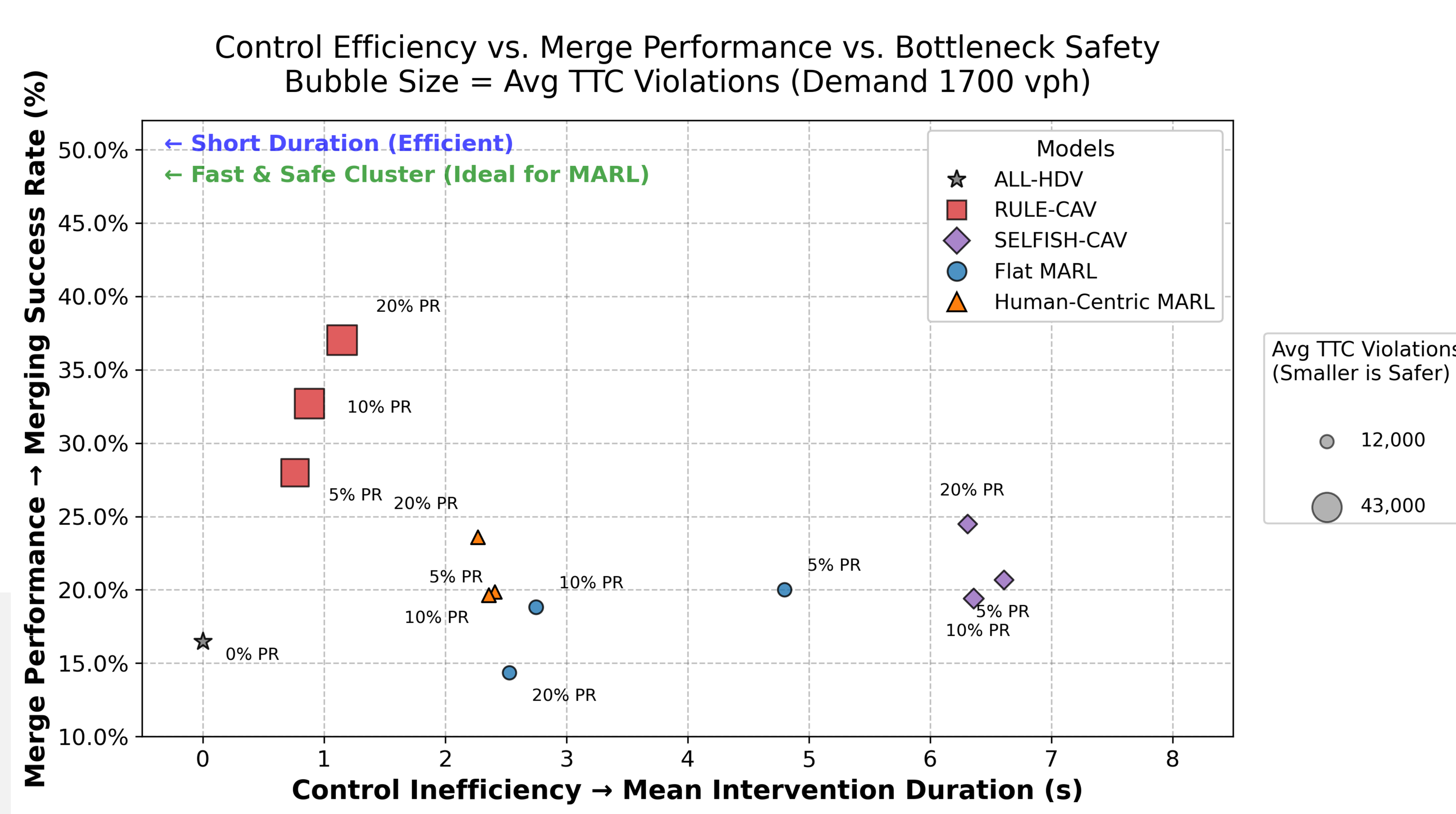


- RULE-CAV recorded a noticeably higher average jerk values, signifying severe ride discomfort.
- Human-Centric MARL achieved an 84.0% better passenger comfort score compared to RULE-CAV and is comparable to the other baselines.
- Human-Centric MARL also achieved very similar throughput compared to baselines.





- Human-Centric MARL recorded 69.9% less Time-to-Collision (TTC) violations compared to RULE-CAV in 1700 vph demand conditions.
- At both 800 vph and 1700 vph, Human-Centric MARL averaged an intervention duration of around 2.3 seconds, lower than both Flat-MARL and SELFISH-CAV.
- Compared against the ALL-HDV baseline, the Human-Centric MARL achieved higher overall merging success rate.



Conclusion

- Overall, the Human-Centric MARL maintains strong traffic efficiency, collision safety, and comfort metrics in both low and high demand conditions.
- It delivers these results using significantly shorter intervention windows that respect human driving dynamics, making it more viable for real-world usage.
- More training and testing is still on-going as more data is collected and analyzed.

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References

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